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When Does Latent Prediction Error Support Physical Surprise Claims in Video World Models?

When is a video world model’s latent prediction error “physical surprise”?

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In brief

Physics is partly decodable and actions are used, yet native error mainly tracks **trajectory type, color and static content**. In controls, **impossible events can get lower error than possible ones**, so it can’t be read as physical surprise.

Code & data log: github.com/3-ml/jepa-surprise

High prediction error need not imply a physics violation

Prediction-error score

$$S(x, a) = d(p(f_C(x), a), f_T(x))$$

f_C / f_T: context and target representations (incl. encoding and token selection)

p: predictor; **a**: optional action; **d**: scoring distance

It measures: **how hard prediction is in this representation**

Common uses

- Violation-of-expectation (VoE) detection
- IntPhys and other physics benchmarks
- Physics scoring / reward for generated video

Higher error may come from…

Physics violation	What we actually want to measure
Appearance	Color and brightness changes
Static content	What is in the scene
Motion	More motion, harder prediction
Ordinary difficulty	The model’ s own error

Method

Framework: separate “what’s encoded” from “what error tracks”

Readout

Repr.

Train linear probes on frozen features to separate possible/impossible events in unseen scenes

Geometry

Repr.

Per layer: relative size, directional consistency and subspace overlap of appearance/physics shifts

Action

Error

Fix the observed outcome; compare correct vs. wrong actions to see if error rises

Appearance / static

Error

Change only color (same physical state) or repeat one frame as a clip; compare with the original

Physical validity

Error

Do impossible events score higher error under every motion/appearance condition?

Calibration

Error

Set a threshold on calibration data, transfer it unchanged to new scenes, count false alarms and misses

Models evaluated: V-JEPA2-G (IntPhys2, Kubric) • V-JEPA2-L (Physion++) • VideoMAE2-G and pixel statistics (controls) • LeWorldModel / DINO-WM / JEPA-WM (TwoRoom, PushT). Each model keeps its official pipeline; errors are compared only within the same model.

Key design: same trajectory, opposite validity in two worlds

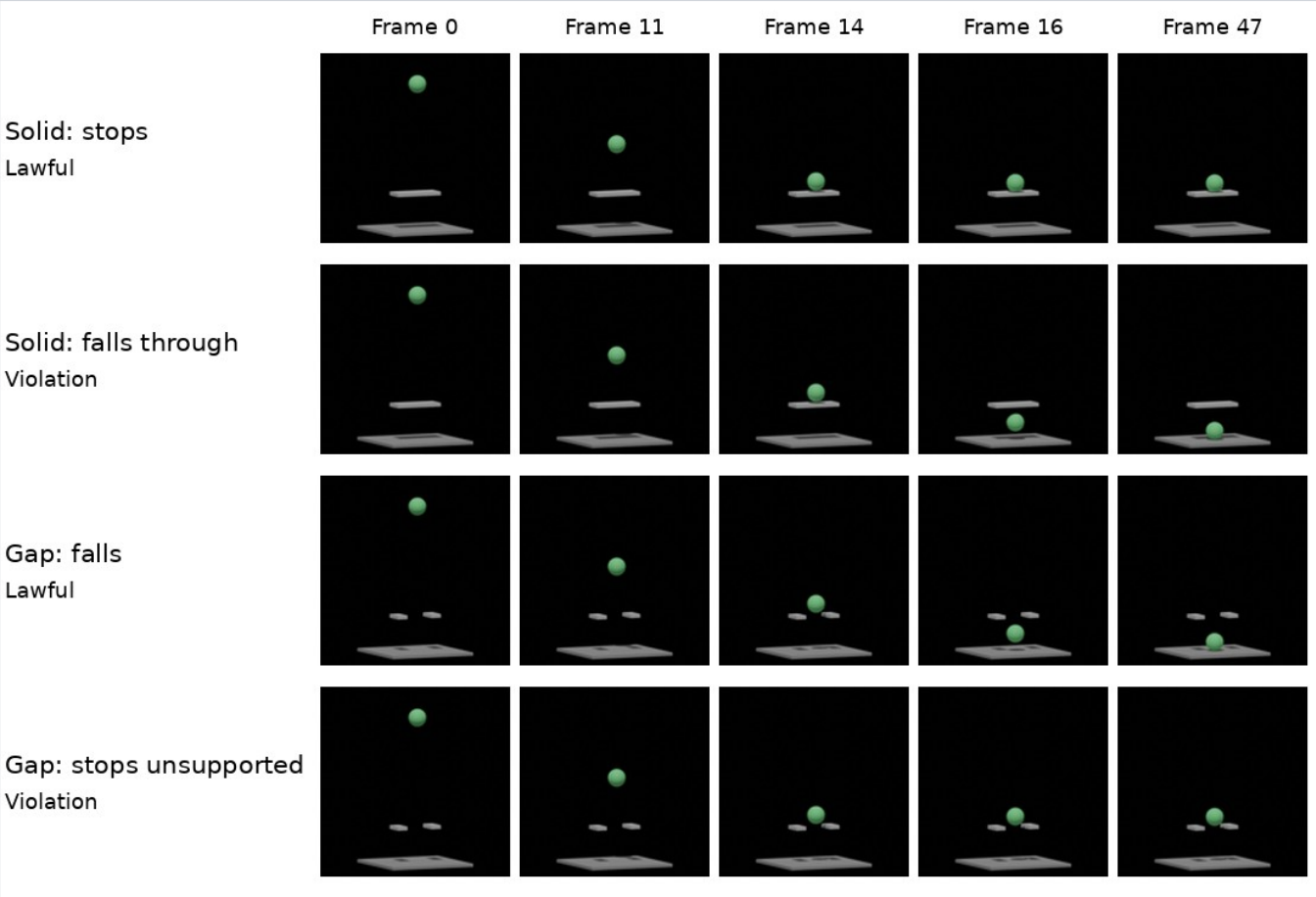


Fig. 8: middle platform is solid or has a central gap; a lower platform catches the falling ball

	Stops at platform height	Keeps falling
Solid platform	Valid	**Violation** Passes through solid
With gap	**Violation** Hovers unsupported	Valid

256 configs × 4 cases × 3 appearances = **3,072** clips

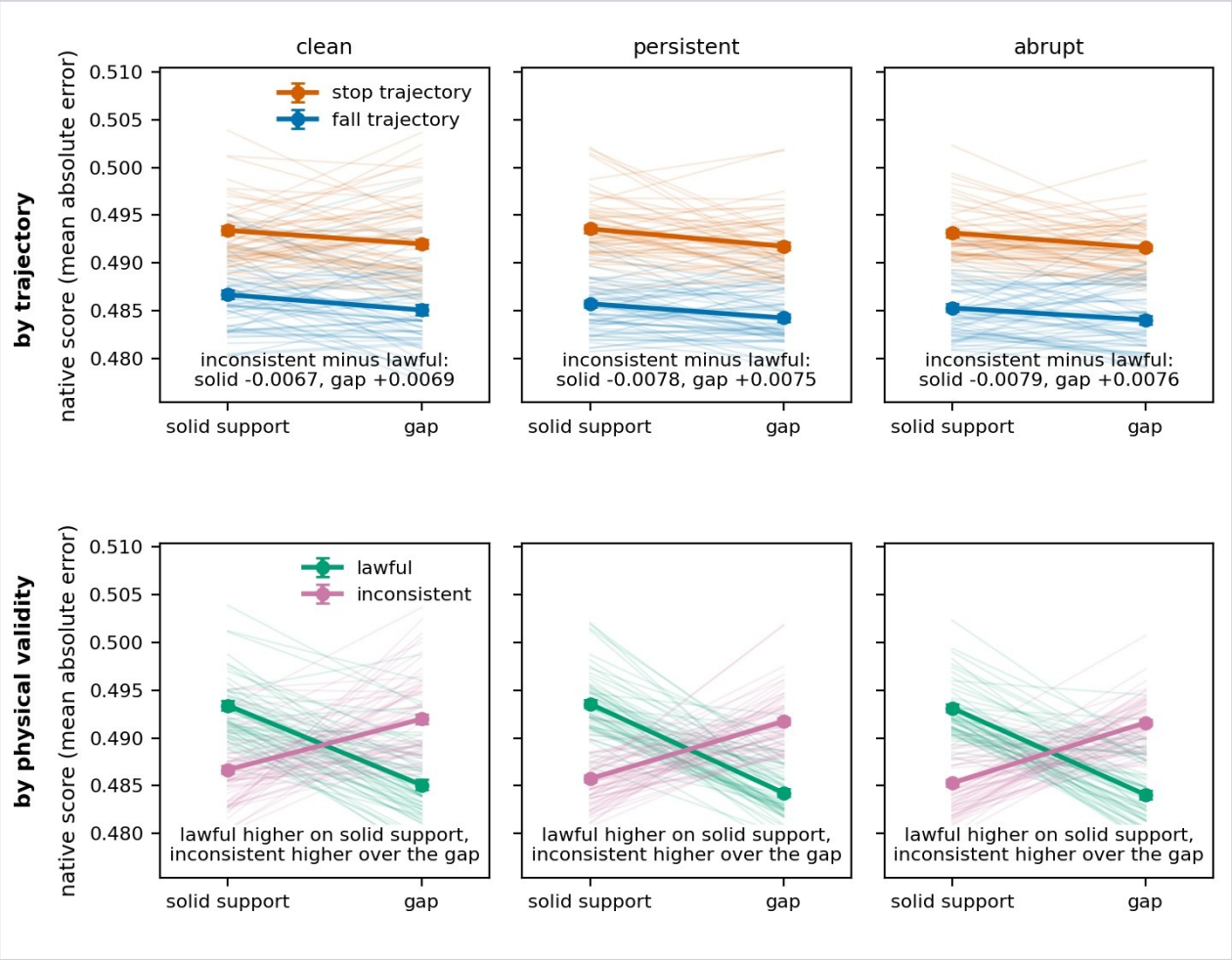
Appearance: original / full-clip 30° hue / shift from target onset

V-JEPA2-G sees 12 frames, predicts the next 36; error = MAE vs. EMA target tokens

"Stop" trajectories are identical across worlds; so are "fall" trajectories

Core result 1

Error follows trajectory, not physical validity



Impossible − possible (clean)

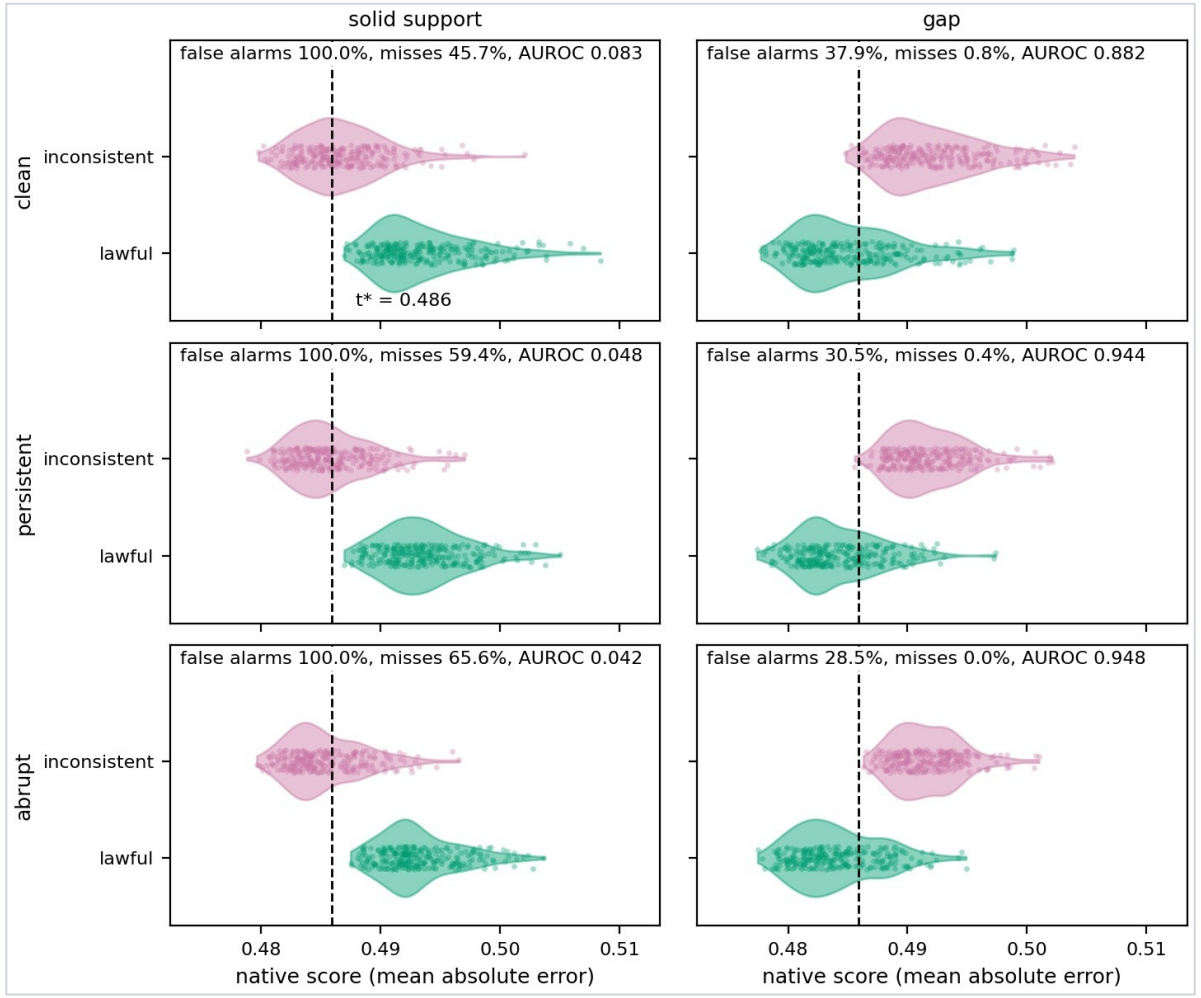
−0.0067
Solid:
violations lower

+0.0069
With gap
Violations score higher

- In both worlds, **"stop" trajectories have higher error** (≈0.493 vs. 0.485)
- So the sign of the violation gap flips with support; under clean and full-hue appearances, all 4 corrected intervals exclude 0
- Averaging over both worlds cancels the opposite orderings, **masking the problem**

Core result 2

Fixed threshold: 100% of valid solid-platform clips flagged



53.9%
Clean balanced acc.
(chance = 50%)

~69%
Valid clips falsely flagged

27.1%
Solid-platform acc.
AUROC 0.083

80.7%
Gap accuracy
AUROC 0.882

- Threshold t* = 0.486, fit on 24 held-out clean configs; no refitting, no direction flip
- AUROC near 0 on solid platforms: score direction is almost fully inverted
- A “slightly above chance” average hides many false alarms

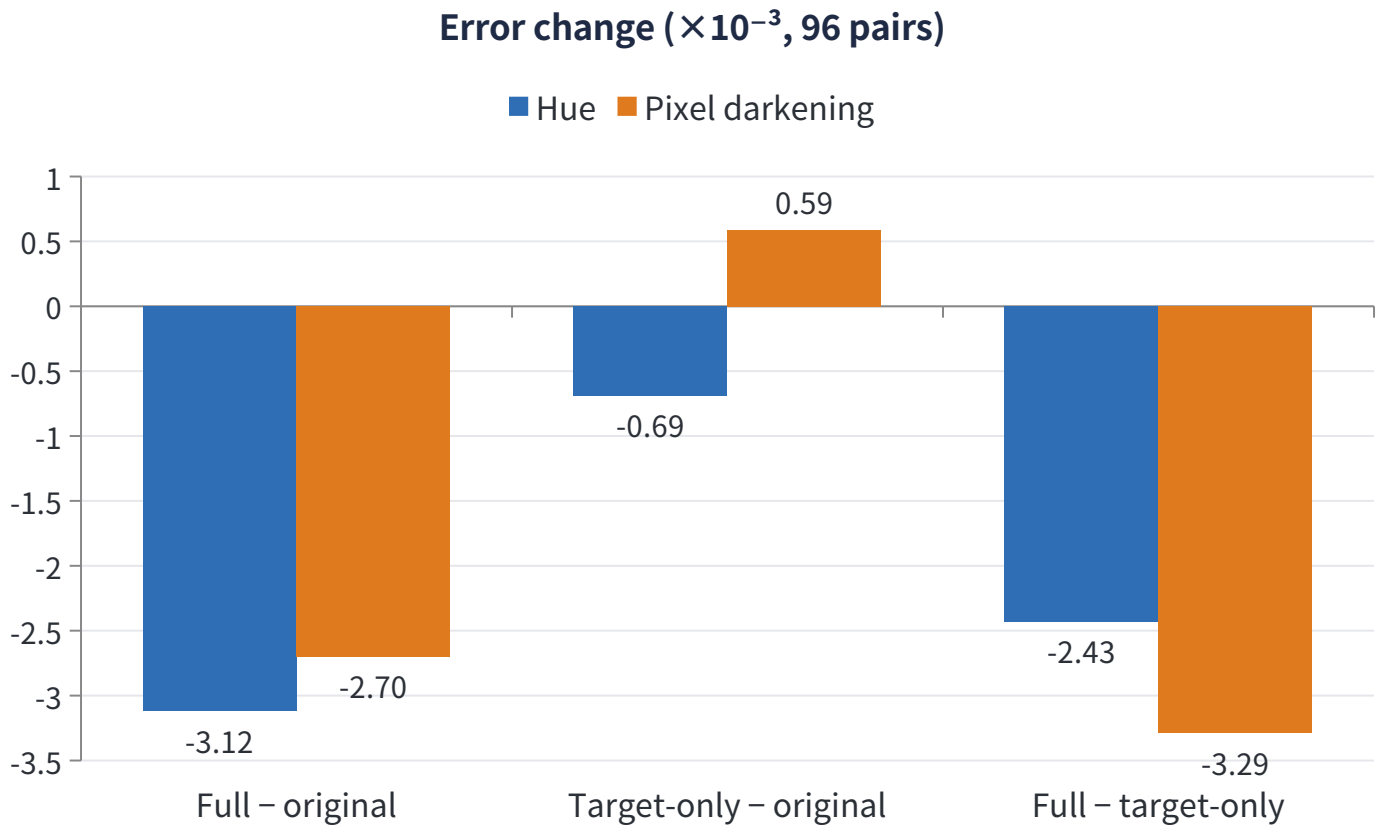
Removing motion raises error; full recoloring lowers it

≈84%

Original window error is no higher than the "repeated-frame" static version (V-JEPA2-L)

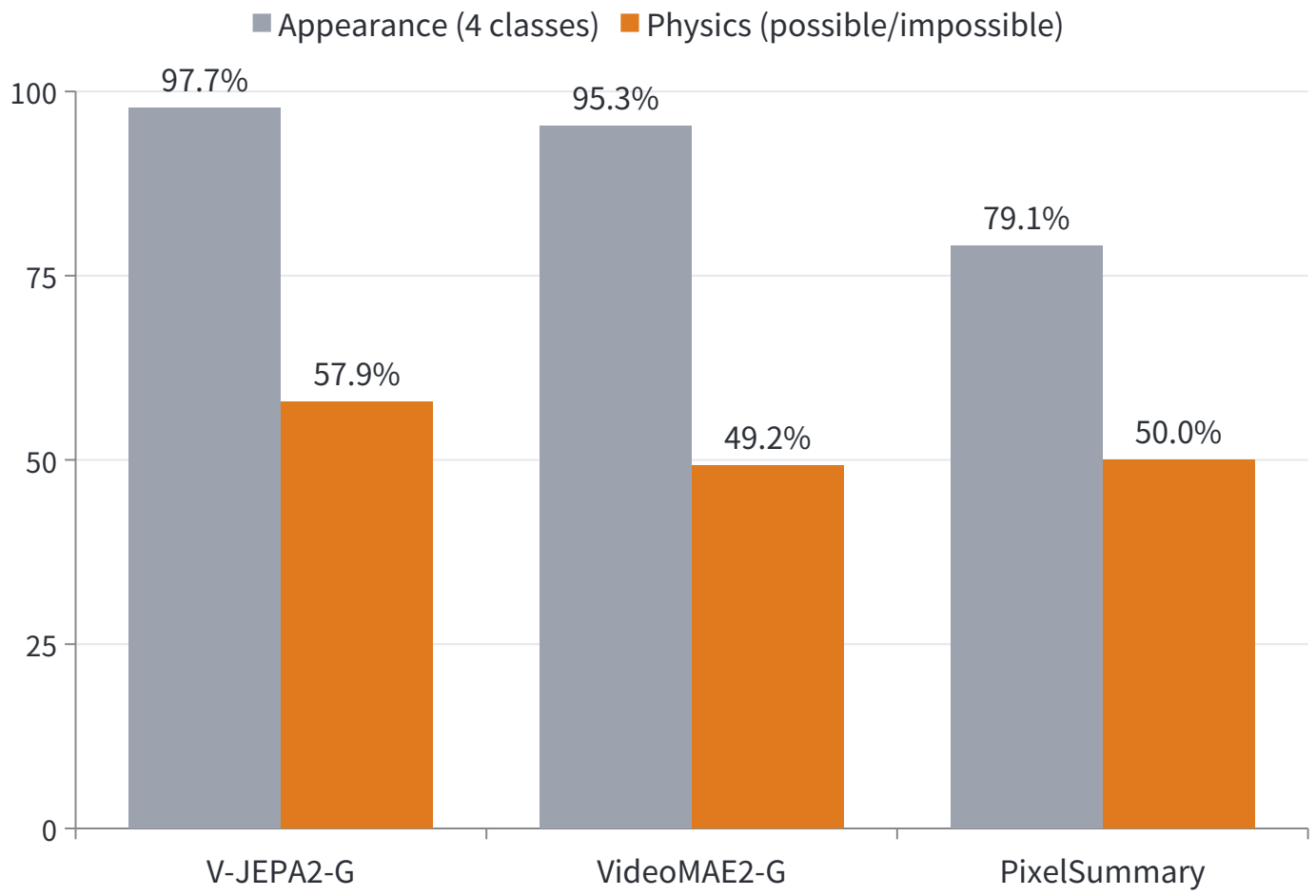
- 192 official Physion++ Mass videos; each full 16-frame window vs. versions repeating the first/middle/last frame
- Mean differences are negative in all three scenarios

Note: the official code encodes all 16 frames before selecting context tokens, leaking target-frame information to the predictor; conclusions apply only to this implementation



Both full-recolor intervals exclude 0; both "target-only" intervals span 0. Data: Table 2 (99.17% paired bootstrap intervals)

Physics is decodable but weak; appearance almost fully so



+8.7 pp

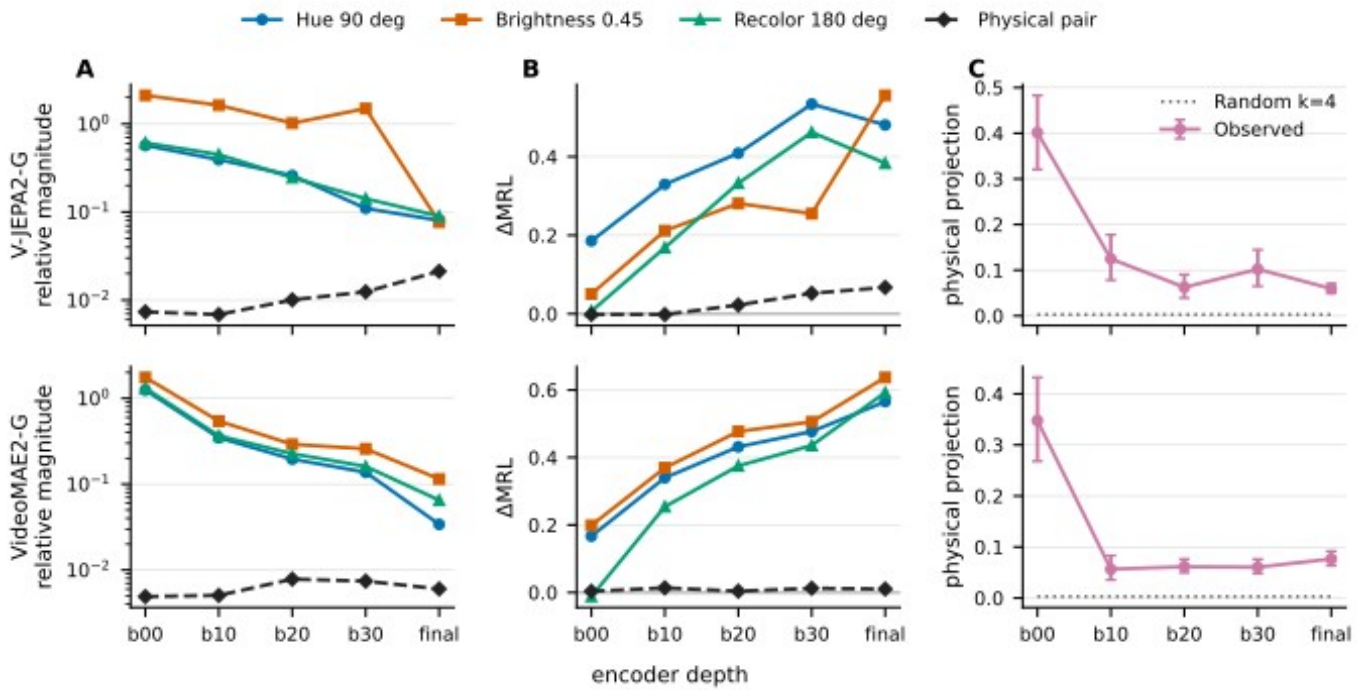
V-JEPA2-G physics-probe gain over VideoMAE2-G
CI [4.4, 13.3]

+7.9 pp

V-JEPA2-G physics-probe gain over pixel stats
CI [4.1, 11.8]

- 57 test scenes; 5-fold scene-wise cross-fitting; class-balanced logistic regression
- With clean clips only, V-JEPA2-G's edge over pixel statistics is not significant
- Architectures and training differ, so gains can't be attributed to the JEPA objective; the analysis is post hoc

Deeper layers: appearance shifts shrink, align, overlap physics less



A Relative magnitude

57% → 8%

V-JEPA2-G hue shift / median distance between unrelated videos (first → last layer)

B Directional consistency ΔMRL

0.4–0.6 vs ≈0.07

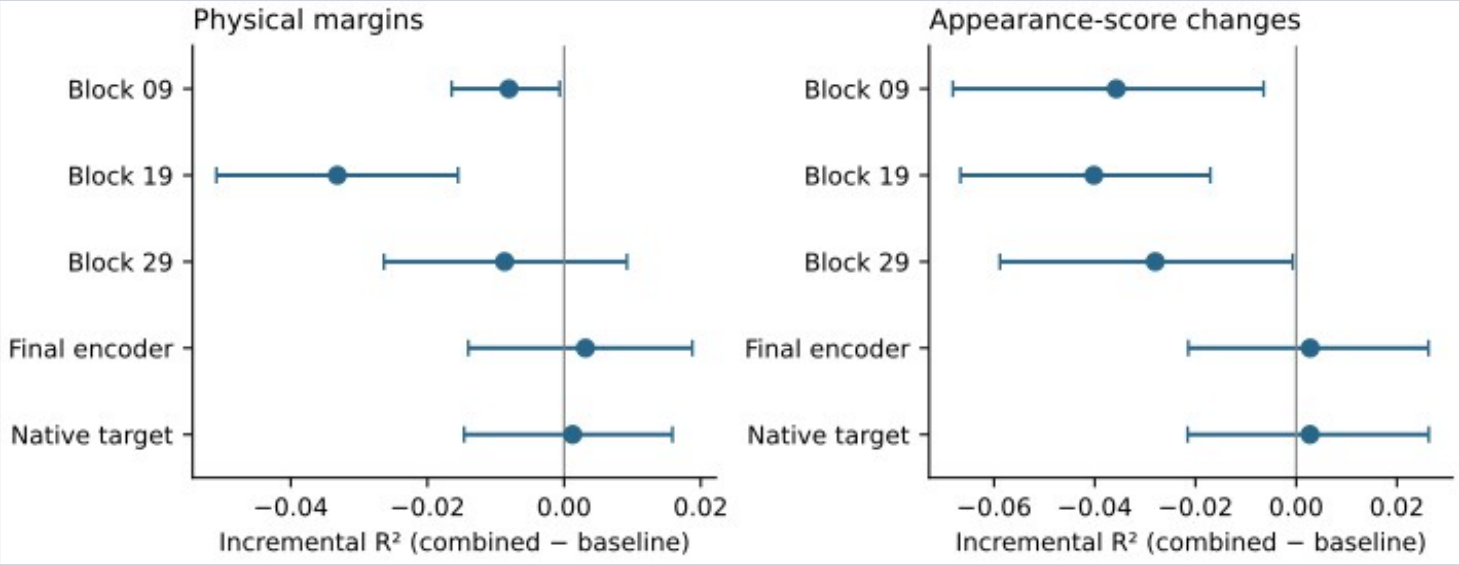
Deep appearance shifts are aligned; physics shifts stay near the shuffled baseline

C Appearance subspace overlap

0.40 → 0.06

Still above random baseline 0.0028; decline is non-monotonic (b30 rises to 0.10)

Elegant geometry barely explains native error differences



0.771 → 0.774

Physics-margin R^2: with last-layer geometry, the increment's interval spans 0

0 / 10

Of 5 layers × 2 tasks, cases with increment interval fully above 0

- Baseline: condition labels plus static, motion and shuffled-frame statistics; extended model adds representation distances and interactions
- Block 19 is significantly worse (−0.033); for appearance score changes, even the baseline R^2 is negative

Fig. 12: incremental R^2 from adding geometric features (99.5% config-level bootstrap intervals)

Action-conditioned models use actions but react more to color

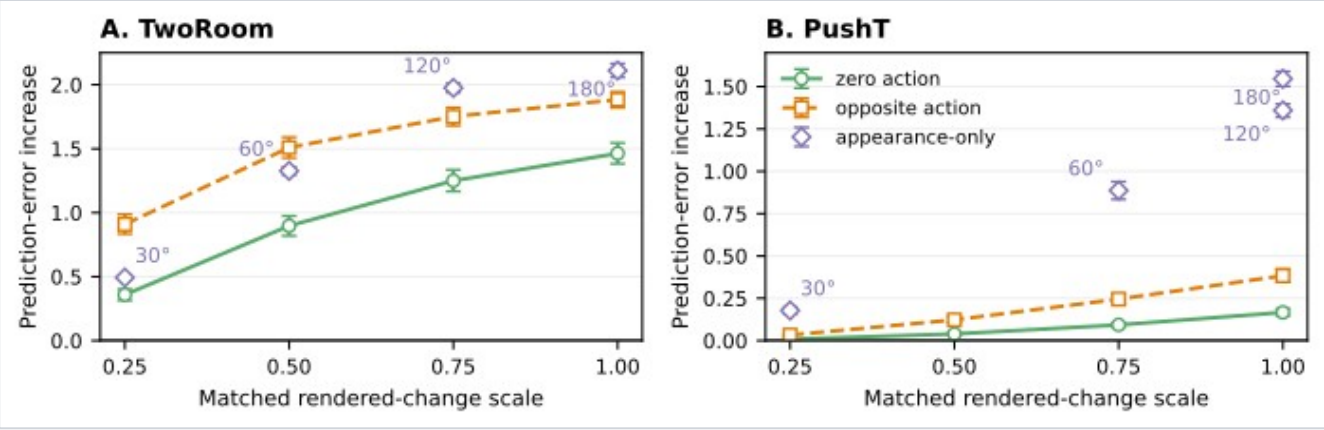


Fig. 3: LeWM error increments; purple diamonds = color change only (same physical state)

Actions matter: with the target frame fixed, opposite actions raise error (LeWM TwoRoom +1.88, PushT +0.38; also positive for DINO-WM and JEPA-WM)

PushT: hue 60° vs. action scale 0.75 increments

Model	Hue Δ	Action Δ	Ratio
LeWM	0.887	0.245	≈3.6×
DINO-WM	0.0498	0.0031	≈16×
JEPA-WM	0.0174	0.0028	≈6×

Δ vs. own valid baseline; ratio from Table 15 means, descriptive

Baseline trap

For DINO-family models, comparing raw scores makes hue appear "lower" than opposite actions, because the valid baseline for moving targets is already much harder (−0.157 / −0.139). The paper isolates this term with a decomposition.

Error depends on which color, not just how much it changes

Physical state unchanged, RGB change magnitude strictly matched: block painted agent blue vs. novel orange; agent painted block gray vs. novel purple

+0.493

Block: blue−orange (target only)
233/256 episodes positive

+0.397

Agent: gray−purple (target only)
245/256 episodes positive

- With full recoloring, gaps shrink (0.111 / 0.088), but both orderings persist
- Exact decomposition: recoloring only target frames leaves predictions unchanged, so the score gap ≈ the **target-encoder shift gap** (0.493 vs. 0.495), i.e., the encoder's sensitivity to color direction
- Color itself is informative in the task; this experiment **cannot** show whether LeWM encodes "roles"

Summary

Neither readout nor action use shows “error = physical surprise”

Model / data	Readout	Geometry	Action	Appear./static	Physical validity	Threshold transfer
V-JEPA2-G / IntPhys2	Pass	Pass	–	–	–	–
VideoMAE2-G / IntPhys2	Weak	Pass	–	–	–	–
V-JEPA2-G / cross-support	–	–	–	Depends	Conflict	Weak
V-JEPA2-L / Physion++	–	–	–	Depends	–	–
LeWM / TwoRoom	–	–	Pass	Depends	Weak	–
LeWM / PushT	–	–	Pass	Depends	Weak	–
DINO-WM / PushT	–	–	Pass	Depends	Weak	–
JEPA-WM / PushT	–	–	Pass	Depends	Weak	–

Pass = check passed; **Weak** = insufficient evidence; **Depends** = error affected by appearance/static content; **Conflict** = on solid platforms, impossible events have lower mean error than possible ones; – = not tested

Takeaway: two tests before calling error “physical surprise”

Recommendations for users

1

Within-condition violation ranking

Within each relevant condition (support, appearance, motion), do violations reliably score higher? Averages alone are not enough

2

Fixed-threshold transfer

False-alarm and miss rates after transferring the threshold unchanged to new scenes and appearances

3

Before use as reward or safety signal

Re-validate in the target task and environment to learn what the error actually responds to

Limitations (acknowledged)

- Cross-support covers one procedural scene type (single ball; limited camera/color/lighting); no real videos tested
- IntPhys2 readout and geometry→error regression are post hoc; intervals omit refitting uncertainty
- Physion++ official scoring leaks target-frame information; Physion++ has no violation labels
- Models differ in architecture, scale and pretraining, so no attribution to training objectives; the paper is diagnostic and proposes no fix